COMMUNITY DISASTER VOLUNTEER HUB

CDVH 01

# Comprehensive Disaster Management Fundamentals

Hazard/domain: multi-hazard
Geographic scope: India, with Kerala-focused community examples

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| **Use limitation** | Educational reference; not a live alert, operational order, professional inspection, medical treatment or legal advice |
| **Research status** | Official source pages and source limitations are listed in this document |

**Safety rule: if a person is in immediate danger or a current evacuation instruction applies, take protective action and contact local emergency services. Do not wait for a chatbot to complete an assessment.**

# Contents and document use

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## How to use this document

Sections are designed to be retrieved independently. Preserve the hazard, geography, time period, conditions and source context when indexing. This PDF is reference material; current warnings, road closures, shelter status and live environmental readings require a current authorized source.

## Section map


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Additional components: 24 FAQs, a glossary, references and source/update limitations.

## Risk concepts

Susceptibility describes relative tendency for a place to experience a process. Hazard concerns the potentially damaging event and its characteristics over a defined period. Exposure concerns people and assets in harm's way. Vulnerability describes conditions that increase susceptibility to harm. Capacity includes resources and abilities that help reduce consequences. Risk concerns potential adverse consequences considering these dimensions and uncertainty. A susceptibility map is not an imminent-event forecast; a low map class is not a guarantee of safety.

# 01 Core science and definitions

Document 01 | Section 01 | Hazard/domain: multi-hazard | Scope: India, with Kerala-focused community examples

Disaster risk is shaped by the interaction of a hazardous process with exposed people and assets, their vulnerability, and their capacities. A hazard is not itself a disaster: consequences depend on place, timing, intensity, preparedness, infrastructure and response capability. Disaster management spans prevention, mitigation, preparedness, response, recovery and learning, rather than only relief after an event.

Use the terms in this section carefully. A hazard description identifies the physical or operational process; it does not by itself establish that a particular household will be affected. For public guidance, name the process, the relevant geography and the time horizon before discussing consequences. In technical subjects, define abbreviations and preserve the unit, scale or jurisdiction that gives a term its meaning.

For a retrieved passage about multi-hazard, the minimum useful context is: what process is being discussed, where the statement applies, whether it is general or current, who issued the information, and what practical decision the reader is trying to make. A passage missing one of these may still explain a concept, but it should not be used alone for an operational decision.

For multi-hazard, apply the section's principle to the actual mechanism, location and time horizon. Record what is observed, what is sourced, what is inferred and what is unknown. Do not extend a general rule beyond its stated scope; use local official instructions for decisions that depend on present conditions.

## Decision aid

| Term | Meaning for decision support | Do not infer |
|---|---|---|
| Hazard / process | What could cause harm and how it may operate | That harm is occurring at a named house now |
| Exposure | People, buildings, services or livelihoods in the affected area | That all exposed people have equal vulnerability |
| Vulnerability / capacity | Conditions that can increase harm or improve coping | That a map or average describes every person |
| Risk statement | Potential consequences over a stated period and context | An exact probability without suitable evidence |

## Practical checks

- Write the geography and time window near the claim.
- Attribute official instructions to the issuing authority and preserve issue time.
- Keep uncertain inference separate from observation and published guidance.

Supporting source identifiers: [S01], [S02], [S07], [S08]. Full titles, publishers, dates, URLs and limitations appear in References and source notes.

# 02 Causes, contributing factors, and mechanisms

Document 01 | Section 02 | Hazard/domain: multi-hazard | Scope: India, with Kerala-focused community examples

Risk drivers include unsafe land use, fragile buildings, degraded ecosystems, poverty, inaccessible services, limited warning reach, weak coordination and climate or environmental changes. The same storm can produce different consequences across neighbouring communities because exposure and vulnerability differ. Capacities such as trusted networks, safe shelters, accessible communications and trained responders reduce expected harm.

A useful explanation separates background susceptibility, a trigger, a pathway of impact and exposed assets. The word 'cause' should be used with care: several conditions may contribute, and the data available to a chatbot may not establish which factor controlled an individual event. The relevant scientific mechanism for multi-hazard must be matched to the actual site or system rather than assumed from a neighbouring district.

When explaining a possible event, show how evidence links each stage. For example, a regional weather event may create a broad loading or stress condition, but local terrain, drainage, construction, infrastructure condition or exposure can change the result. In other topics, the same logic applies to river-network routing, coastal water levels, building response, fire spread or gaps in a technical data pipeline. A plausible pathway is not proof that the pathway will occur.

For multi-hazard, apply the section's principle to the actual mechanism, location and time horizon. Record what is observed, what is sourced, what is inferred and what is unknown. Do not extend a general rule beyond its stated scope; use local official instructions for decisions that depend on present conditions.

## Decision aid

| Stage | Question to ask | Evidence limitation |
|---|---|---|
| Precondition | What makes this place or system susceptible? | May be incompletely mapped or not visible |
| Trigger | What recent forcing or failure could initiate change? | A trigger rarely determines the exact outcome alone |
| Pathway | How could the process reach people/assets? | Terrain, network or building connection matters |
| Consequence | Who or what could be affected, and how? | Exposure and protective capacity vary locally |

## Practical checks

- Use 'may contribute' when the evidence is conditional.
- Do not assert a single cause from timing alone.
- Route technical mechanism questions to qualified authorities when site-specific judgment is necessary.

Supporting source identifiers: [S01], [S02], [S07], [S08]. Full titles, publishers, dates, URLs and limitations appear in References and source notes.

# 03 Indicators, warning signs, and limitations

Document 01 | Section 03 | Hazard/domain: multi-hazard | Scope: India, with Kerala-focused community examples

Indicators differ by hazard. Examples include an official warning, rapidly rising water, new slope cracks or seepage, strong shaking, smoke, extreme heat symptoms or a coastal natural tsunami warning. A single indicator may be uncertain, but serious direct signs require protective action. Do not wait for a complete checklist when life may be threatened.

Warning signs are observations that can justify protective action or reporting; they are not a complete diagnostic test. The evidence may be early, ambiguous or absent. A checklist cannot promise that an event will be detected before it becomes dangerous. Users should observe from a safe location and report what they saw, where and when, without approaching the hazard to obtain a clearer picture.

The safest public workflow is to separate direct observation from a forwarded message, a model interpretation and an official bulletin. Record the time, approximate place, direction of change and whether the observation is continuing. If the sign suggests immediate danger, first move away by a safe route and alert appropriate emergency services; the chatbot should not ask for photos or multiple confirmations before recommending that action.

For multi-hazard, apply the section's principle to the actual mechanism, location and time horizon. Record what is observed, what is sourced, what is inferred and what is unknown. Do not extend a general rule beyond its stated scope; use local official instructions for decisions that depend on present conditions.

## Decision aid

| Evidence type | Useful interpretation | Safe next action |
|---|---|---|
| Direct, changing physical sign | May indicate a developing local danger | Move away if threatened; report from safety |
| Official alert or order | Authoritative for its stated area/time and instruction | Follow it; verify updates as appropriate |
| Third-party message / screenshot | Unverified until source and timestamp are checked | Do not forward as fact; verify through official channel |
| No visible sign / no alert | Absence of evidence, not proof of safety | Use other relevant evidence and current guidance |

## Practical checks

- Never enter, cross or approach a hazard to verify a sign.
- Do not require every listed sign to be present.
- Explain when signs are nonspecific and why qualified assessment may be needed.

Supporting source identifiers: [S01], [S02], [S07], [S08]. Full titles, publishers, dates, URLs and limitations appear in References and source notes.

# 04 Location-specific risk assessment

Document 01 | Section 04 | Hazard/domain: multi-hazard | Scope: India, with Kerala-focused community examples

Start with the event, place, time window and current official warnings. Separate hazard likelihood and characteristics from exposure, vulnerability and coping capacity. Use maps as screening tools, not individual guarantees. Record the source, timestamp, spatial coverage, assumptions and unknowns for every risk statement. A qualitative category is preferable to an invented percentage.

A preliminary risk screen should answer four distinct questions: what is the hazard or process; what present evidence describes it; which people or assets may be exposed; and what conditions make the consequences more or less severe. Static maps and regional alerts are inputs, not individual guarantees. State the observation date, map scale or forecast period, geographic coverage and uncertainty whenever known.

For a location-specific question about multi-hazard, compare the spatial and operational relationship that actually controls the hazard, rather than using straight-line distance alone. State the time window, the relevant conditions, the freshest applicable official source and the main unknowns. If those details are unavailable, describe what can be concluded qualitatively and what would require a site-specific professional assessment.

For multi-hazard, apply the section's principle to the actual mechanism, location and time horizon. Record what is observed, what is sourced, what is inferred and what is unknown. Do not extend a general rule beyond its stated scope; use local official instructions for decisions that depend on present conditions.

## Decision aid

| Evidence | What it can support | What it cannot establish alone |
|---|---|---|
| Susceptibility / hazard map | Relative spatial screening under a method and scale | Exact event time or guaranteed household safety |
| Current official warning | Action and forecast for its stated validity/area | Conditions beyond its coverage or after expiry |
| Local observation | A potentially important current sign | Complete scientific diagnosis or probability |
| Validated local model | Quantitative estimate if model is fit for task | Certainty outside training/validation conditions |

## Practical checks

- Do not invent risk percentages from distance or a map color.
- Compare the source timestamp and locality before combining evidence.
- If immediate danger is reported, give protective action before optional assessment questions.

Supporting source identifiers: [S02], [S07], [S08], [S09]. Full titles, publishers, dates, URLs and limitations appear in References and source notes.

# 05 Prevention and mitigation

Document 01 | Section 05 | Hazard/domain: multi-hazard | Scope: India, with Kerala-focused community examples

Prevention aims to avoid creating new risk; mitigation reduces existing risk. Examples include safer siting, code-compliant construction, drainage maintenance, ecosystem protection, access to warning channels, safe electrical design and reducing barriers to evacuation. Measures must suit the hazard and local evidence; a structure or intervention can transfer risk if poorly designed.

Mitigation is most effective when it changes the actual pathway to harm rather than merely improving the appearance of safety. Options may include land-use controls, maintenance, resilient design, drainage or vegetation management, monitoring, warning access and safer work practices, depending on the hazard. Measures that require engineering, geotechnical, structural, electrical, medical or legal judgment must be designed or approved by qualified people.

For multi-hazard, choose measures that address the mechanism documented by the responsible authority and consider maintenance, unintended effects, accessibility and accountability. Record the objective, responsible owner, expected benefit, limitations and review date. Any intervention affecting public infrastructure or life safety needs approval and technical competence appropriate to the task.

For multi-hazard, apply the section's principle to the actual mechanism, location and time horizon. Record what is observed, what is sourced, what is inferred and what is unknown. Do not extend a general rule beyond its stated scope; use local official instructions for decisions that depend on present conditions.

## Decision aid

| Mitigation layer | Purpose | Governance check |
|---|---|---|
| Avoid exposure | Keep people/assets out of known dangerous pathways | Use current land-use and authority guidance |
| Reduce vulnerability | Improve structures, services or protections | Use applicable code and qualified design |
| Monitor and warn | Provide lead time and actionable messages | Validate system coverage and alert reach |
| Maintain / review | Prevent deterioration and detect new conditions | Assign responsibility, budget and review date |

## Practical checks

- Prefer risk reduction that does not transfer danger to neighbours.
- Do not offer a remote certification of a site, structure or utility.
- Keep technical intervention separate from low-risk household actions.

Supporting source identifiers: [S02], [S07], [S08], [S09]. Full titles, publishers, dates, URLs and limitations appear in References and source notes.

# 06 Preparedness and readiness

Document 01 | Section 06 | Hazard/domain: multi-hazard | Scope: India, with Kerala-focused community examples

Maintain a household and community plan with evacuation routes, meeting points, contacts, medicines, essential documents, water, lighting, charging options and support arrangements for people needing assistance. Assign roles, test communications, identify alternate routes and conduct drills. Review plans before high-risk seasons and after changes in household or neighbourhood conditions.

Preparedness works only when it can be performed under adverse conditions: darkness, rain, power loss, network outage, road disruption, caregiving responsibilities and limited cash or transport. A household or community plan should identify who receives warnings, who needs help, where to move, what to carry and how to reunite if devices fail. Keep essentials accessible rather than locked behind a single person or device.

Before the relevant season or event, confirm contact details, alert subscriptions, household medications and support needs, alternate routes, transport options and shelter information through an official source. Do not assume that a shelter listed in an old plan is open today. Practice roles and update them when a road, residence, caregiver or mobility need changes.

For multi-hazard, apply the section's principle to the actual mechanism, location and time horizon. Record what is observed, what is sourced, what is inferred and what is unknown. Do not extend a general rule beyond its stated scope; use local official instructions for decisions that depend on present conditions.

## Decision aid

| Time | Readiness action | Verification |
|---|---|---|
| Before the risk season | Check supplies, contacts, routes and support roles | Use current authority/shelter information |
| At warning stage | Charge devices, monitor updates and prepare to move | Confirm geographic coverage and issue time |
| When instructed to act | Take the stated protective action promptly | Do not wait for optional chatbot analysis |
| After an exercise/event | Record what failed and revise the plan | Set an owner and due date for each fix |

## Practical checks

- Make the plan accessible to children, older adults and people with disabilities.
- Include a no-network communication method.
- Keep sensitive medical or identity details private; share only what is needed for assistance.

Supporting source identifiers: [S02], [S07], [S08], [S09]. Full titles, publishers, dates, URLs and limitations appear in References and source notes.

# 07 Immediate response procedures

Document 01 | Section 07 | Hazard/domain: multi-hazard | Scope: India, with Kerala-focused community examples

In an active emergency, protect life first: heed official warnings, move away from immediate danger when safe, call verified emergency services and avoid entering hazard zones. Keep messages concise, identify location and observed condition, and do not spread unverified claims. Responders should coordinate through the designated incident-management structure and track people, tasks and changing hazards.

Response advice must be ordered by urgency. Protect life and avoid turning a bystander into another casualty; use the appropriate emergency channel; follow current official instructions; communicate location and observations from safety; and leave technical tasks to qualified responders. The chatbot should not claim it dispatched help, checked live conditions or sent an alert unless an authorized integration confirms that action.

When a user reports a rapidly worsening condition, keep the initial response short and direct. Give the first safe action before explaining scientific background or asking for a detailed location. A broad location can help identify the relevant authority, but exact coordinates should not be required to recommend avoiding an immediate hazard. Once the person is safe, gather only additional details that affect the next decision.

For multi-hazard, apply the section's principle to the actual mechanism, location and time horizon. Record what is observed, what is sourced, what is inferred and what is unknown. Do not extend a general rule beyond its stated scope; use local official instructions for decisions that depend on present conditions.

## Decision aid

| Priority | Action | Boundary |
|---|---|---|
| 1. Protect | Move away from immediate danger by a safe route | Do not cross the hazard or retrieve belongings |
| 2. Alert | Contact current local emergency/response channels | Do not imply the chatbot called on the user's behalf |
| 3. Inform | Give approximate location, time and observed condition | Report facts separately from assumptions |
| 4. Reassess | Follow responders and current updates | Do not return or begin technical rescue without clearance |

## Practical checks

- Do not delay action to complete a risk score.
- Do not enter unstable slopes, floodwater, smoke, rubble or electrically unsafe areas.
- If the route is compromised, report the obstacle from safety and follow authorized instructions.

Supporting source identifiers: [S02], [S07], [S08], [S09]. Full titles, publishers, dates, URLs and limitations appear in References and source notes.

# 08 Evacuation and protective actions

Document 01 | Section 08 | Hazard/domain: multi-hazard | Scope: India, with Kerala-focused community examples

Evacuation decisions depend on the hazard, route safety, official instructions, timing, mobility and shelter availability. Leave early when ordered or when immediate indicators suggest danger; do not cross floodwater, unstable slopes, fire fronts or damaged structures. If the planned route is compromised, follow official alternate-route guidance instead of improvising into the hazard.

Evacuation depends on timing, route condition, destination and the person's ability to move. A familiar route can become dangerous because of the same event that triggered the evacuation. Use current official route instructions and never infer that a route is safe because it was usable yesterday. If movement is ordered, arrange assistance early and do not wait for a perfect checklist when a threat is immediate.

A decision to shelter in place is hazard-specific and should not be invented by the chatbot. It may be safer for some threats and unsafe for others. The answer must distinguish an official instruction to evacuate, an instruction to remain indoors, a blocked route and a person who sees immediate danger. If an official destination is inaccessible, contact the responsible authority rather than improvising through a dangerous area.

For multi-hazard, apply the section's principle to the actual mechanism, location and time horizon. Record what is observed, what is sourced, what is inferred and what is unknown. Do not extend a general rule beyond its stated scope; use local official instructions for decisions that depend on present conditions.

## Decision aid

| Situation | Safer response principle | Do not |
|---|---|---|
| Explicit evacuation order | Follow it using the stated route and timing | Delay to pack nonessential items |
| Immediate visible danger | Move away if a safe path exists and alert responders | Wait for a model or all warning signs |
| Route appears blocked | Stop before entering the hazard and report it | Cross floodwater or unstable ground |
| All-clear not received | Remain cautious and follow official updates | Return because conditions look calm |

## Practical checks

- Plan for essential medicines, mobility aids, children, older adults and pets where relevant.
- Keep family meeting and reunification arrangements simple.
- Do not promise that a shelter has space without current confirmation.

Supporting source identifiers: [S02], [S07], [S08], [S09]. Full titles, publishers, dates, URLs and limitations appear in References and source notes.

# 09 Volunteer and responder coordination

Document 01 | Section 09 | Hazard/domain: multi-hazard | Scope: India, with Kerala-focused community examples

Community volunteers support only assigned tasks within training, equipment and command arrangements. Log deployment, check in and out, use buddy procedures, communicate hazards, and withdraw when conditions exceed limits. Volunteers should not self-deploy to unstable, contaminated, electrically unsafe or structurally damaged locations, even when trying to help acquaintances.

Community volunteers are most effective when they fill defined support roles and remain within training, equipment and supervision. Proximity to an incident, local familiarity or good intentions do not confer technical rescue capability. Each task needs an assigning supervisor, boundaries, communication method, buddy or accountability procedure, stop-work trigger and safe withdrawal plan.

Examples of safer support may include relaying verified messages, helping an authorized distribution point, translating warnings, assisting with registration under privacy rules, checking on neighbours from safe locations and reporting obstructions. Unsafe self-assigned activities include entering unstable structures, approaching an active landslide, wading into floodwater, going near live wires or attempting rescue from fire, smoke or chemical release. Escalate specialist work to trained responders.

For multi-hazard, apply the section's principle to the actual mechanism, location and time horizon. Record what is observed, what is sourced, what is inferred and what is unknown. Do not extend a general rule beyond its stated scope; use local official instructions for decisions that depend on present conditions.

## Decision aid

| Task type | Volunteer boundary | Stop / withdraw when |
|---|---|---|
| Information relay | Repeat authorized messages with timestamp and area | Message is unverified or outside assigned role |
| Logistics / welfare | Work at an authorized safe location | Crowding, contamination or scene hazard increases |
| Observation / reporting | Observe from safety; report facts and time | Closer approach is needed to get more evidence |
| Technical rescue | Only trained, equipped, authorized teams | Any task exceeds training, command or PPE |

## Practical checks

- Check in, remain accountable and report back before leaving an assignment.
- Use clear identification without impersonating an official.
- Keep emergency access routes clear and protect affected people's privacy.

Supporting source identifiers: [S01], [S02], [S07]. Full titles, publishers, dates, URLs and limitations appear in References and source notes.

# 10 Vulnerable populations and inclusive planning

Document 01 | Section 10 | Hazard/domain: multi-hazard | Scope: India, with Kerala-focused community examples

Preparedness must account for mobility, sensory, cognitive and communication needs; pregnancy; age; language; chronic conditions; and uneven access to transport, phones or cash. Plan with affected people, not just for them. Use accessible formats, trusted interpreters, private consent-based support and safe options for assistance animals and caregivers.

Inclusive risk communication must fit the person, not just the average user. An alert is ineffective if it cannot be heard, seen, understood, reached in time or acted upon. Plan with people who have mobility, sensory, cognitive, language, health, caregiving or transport needs. Avoid assumptions based on age or diagnosis; ask what practical help is needed and preserve the person's agency and privacy.

For India, with Kerala-focused community examples, plan several ways to communicate and move: audible and visual messages, plain language, local-language translations, trusted community networks, accessible transport, caregivers, mobility aids, backup charging and alternatives for people without smartphones. The plan should consider medication continuity and service animals where applicable, without asking users to disclose unnecessary private health information in a public chatbot.

For multi-hazard, apply the section's principle to the actual mechanism, location and time horizon. Record what is observed, what is sourced, what is inferred and what is unknown. Do not extend a general rule beyond its stated scope; use local official instructions for decisions that depend on present conditions.

## Decision aid

| Need / barrier | Planning measure | Check |
|---|---|---|
| Mobility or transport | Accessible transport, assistance role and a route without avoidable barriers | Can the person use the route in time? |
| Hearing / vision | Text, audio, visual and person-to-person warning options | Is the message perceivable and understandable? |
| Language / literacy | Plain language, local translation, tested instructions | Can the person explain the action back? |
| Health / caregiving | Medicine continuity and caregiver coordination | Can essential support travel with the person? |

## Practical checks

- Offer choices and confirm consent when it is possible.
- Do not publish home coordinates, diagnosis or beneficiary data openly.
- Treat inaccessible warning reach as a system failure, not user failure.

Supporting source identifiers: [S01], [S02], [S07]. Full titles, publishers, dates, URLs and limitations appear in References and source notes.

# 11 Recovery and lessons learned

Document 01 | Section 11 | Hazard/domain: multi-hazard | Scope: India, with Kerala-focused community examples

Recovery includes safe access to housing, health care, livelihoods, education, water, sanitation, mental-health support, records and infrastructure. Document damage without entering unsafe sites. Relief allocation should use clear criteria, protect privacy, offer feedback routes and include people who are often missed. Rebuilding should reduce risk rather than recreate pre-disaster exposure.

Recovery should move from immediate relief to safe restoration, rehabilitation and longer-term risk reduction. A cleaned room, reopened road or absent smoke does not prove that a structure, slope, water source or utility is safe. Re-entry and reconstruction decisions may require formal inspection, testing, repair standards or permission from the responsible authority. Record which decision is confirmed and which remains provisional.

A fair recovery system documents needs and assistance transparently while protecting privacy. Track who has received support and who has not been reached, distinguish estimated losses from verified assessments, and provide accessible complaint routes. Recovery planning should consider housing, water, sanitation, health, education, livelihoods, environment and psychosocial support together instead of treating each as a separate checklist.

For multi-hazard, apply the section's principle to the actual mechanism, location and time horizon. Record what is observed, what is sourced, what is inferred and what is unknown. Do not extend a general rule beyond its stated scope; use local official instructions for decisions that depend on present conditions.

## Decision aid

| Stage | Main concern | Evidence / safeguard |
|---|---|---|
| Immediate relief | Safe shelter, water, health, information and protection | Current needs and official coordination |
| Damage assessment | Document loss without entering unsafe places | Source, date, assessor and uncertainty |
| Re-entry / repair | Confirm hazards and systems are safe enough | Authorized inspection or testing when required |
| Build back better | Reduce repeated exposure and vulnerability | Qualified design, community input and maintenance |

## Practical checks

- Do not guarantee compensation eligibility or payment timing without current rules.
- Keep survivor records secure and collect only necessary data.
- Review whether rebuilding reduces risk rather than recreating it.

Supporting source identifiers: [S01], [S02], [S07]. Full titles, publishers, dates, URLs and limitations appear in References and source notes.

# 12 Complex scenarios and chatbot guardrails

Document 01 | Section 12 | Hazard/domain: multi-hazard | Scope: India, with Kerala-focused community examples

A heavy-rain warning covers the district while a family sees water entering a neighbouring street. The chatbot should first determine whether water is rising at the user's location and whether official evacuation orders apply, while immediately advising avoidance of floodwater. It should then direct the user to current official updates and ask only for location and immediate danger details that materially alter next steps.

The scenario should be handled with an evidence-first response. Separate what the user directly observed, what the official source says, what the model can infer and what remains unknown. Offer the safest supported next step first; then explain which additional evidence would materially change the assessment. Never convert a similarity to a past case, a nearby incident, a map color, a sensor threshold or a vector score into a guaranteed outcome.

Expected answer: identify whether water is rising where the family is, tell them not to walk or drive through floodwater, and ask them to follow current local instructions. A district warning is relevant but not a precise house-level assessment. Do not delay a move away from immediate danger.

For multi-hazard, apply the section's principle to the actual mechanism, location and time horizon. Record what is observed, what is sourced, what is inferred and what is unknown. Do not extend a general rule beyond its stated scope; use local official instructions for decisions that depend on present conditions.

## Decision aid

| Response element | What a safe answer should include | What it must avoid |
|---|---|---|
| Immediate action | Protective step if danger is plausible now | Waiting for analysis or more signs |
| Known evidence | User observation and current applicable official source | Treating a screenshot or old PDF as live |
| Uncertainty | Specific unknowns and their relevance | Invented probability or false reassurance |
| Follow-up | One detail that materially changes guidance | Unnecessary precise location or private data |

## Practical checks

- State the limit of the available information plainly.
- Use a current official source when time-sensitive information is required and available.
- If no live check occurred, say it was not checked.

Supporting source identifiers: [S01], [S02], [S07]. Full titles, publishers, dates, URLs and limitations appear in References and source notes.

# Frequently asked questions

General guidance only; follow current instructions from responsible authorities.

**Q1. What is the difference between a hazard and a disaster?**

A hazard is a potentially damaging process or event; a disaster occurs when impacts seriously disrupt a community and exceed its ability to cope. Consequences depend on exposure, vulnerability and capacity. Source trail: [S01], [S02], [S07], [S08]

**Q2. Can risk be reduced even when a hazard cannot be prevented?**

Yes. Safer land use, resilient structures, warning systems, accessible evacuation, prepared services and trusted communication can reduce exposure and consequences. Source trail: [S01], [S02], [S07], [S08]

**Q3. Who decides whether to evacuate?**

Follow current instructions from the responsible local authority or emergency service. If immediate physical danger is evident, move away when a safe route exists and contact responders. Source trail: [S01], [S02], [S07], [S08]

**Q4. Does no warning mean there is no danger?**

No. Monitoring gaps, rapidly developing events or local conditions can outpace warnings. Treat credible direct signs seriously and consult current official channels. Source trail: [S01], [S02], [S07], [S08]

**Q5. What are the main factors for assessing multi-hazard risk?**

Start with the event, place, time window and current official warnings. Separate hazard likelihood and characteristics from exposure, vulnerability and coping capacity. Use maps as screening tools, not individual guarantees. Record the source, timestamp, spatial coverage, assumptions and unknowns for every risk statement. A qualitative category is preferable to an invented percentage. Source trail: [S01], [S02], [S07], [S08]

**Q6. Which warning signs matter for multi-hazard?**

Indicators differ by hazard. Examples include an official warning, rapidly rising water, new slope cracks or seepage, strong shaking, smoke, extreme heat symptoms or a coastal natural tsunami warning. A single indicator may be uncertain, but serious direct signs require protective action. Do not wait for a complete checklist when life may be threatened. Source trail: [S01], [S02], [S07], [S08]

# Frequently asked questions (continued)

Document 01 | Questions 7-12

**Q7. What is the main immediate response for multi-hazard?**

In an active emergency, protect life first: heed official warnings, move away from immediate danger when safe, call verified emergency services and avoid entering hazard zones. Keep messages concise, identify location and observed condition, and do not spread unverified claims. Responders should coordinate through the designated incident-management structure and track people, tasks and changing hazards. Source trail: [S02], [S07], [S08], [S09]

**Q8. How should a household prepare for multi-hazard?**

Maintain a household and community plan with evacuation routes, meeting points, contacts, medicines, essential documents, water, lighting, charging options and support arrangements for people needing assistance. Assign roles, test communications, identify alternate routes and conduct drills. Review plans before high-risk seasons and after changes in household or neighbourhood conditions. Source trail: [S02], [S07], [S08], [S09]

**Q9. What can volunteers safely do during multi-hazard?**

Community volunteers support only assigned tasks within training, equipment and command arrangements. Log deployment, check in and out, use buddy procedures, communicate hazards, and withdraw when conditions exceed limits. Volunteers should not self-deploy to unstable, contaminated, electrically unsafe or structurally damaged locations, even when trying to help acquaintances. Source trail: [S02], [S07], [S08], [S09]

**Q10. How should people needing additional assistance be included?**

Preparedness must account for mobility, sensory, cognitive and communication needs; pregnancy; age; language; chronic conditions; and uneven access to transport, phones or cash. Plan with affected people, not just for them. Use accessible formats, trusted interpreters, private consent-based support and safe options for assistance animals and caregivers. Source trail: [S02], [S07], [S08], [S09]

**Q11. What should happen after the event?**

Recovery includes safe access to housing, health care, livelihoods, education, water, sanitation, mental-health support, records and infrastructure. Document damage without entering unsafe sites. Relief allocation should use clear criteria, protect privacy, offer feedback routes and include people who are often missed. Rebuilding should reduce risk rather than recreate pre-disaster exposure. Source trail: [S02], [S07], [S08], [S09]

**Q12. Does distance alone prove safety or danger?**

No. Distance by itself does not establish risk. Evaluate the hazard-specific spatial connection, local conditions, time window, official warning and relevant observations. Source trail: [S02], [S07], [S08], [S09]

# Frequently asked questions (continued)

Document 01 | Questions 13-18

**Q13. Can an exact probability be given from these documents?**

Only if a suitable validated, current and location-relevant model or authoritative forecast explicitly supports it. Otherwise the system must state uncertainty and avoid invented percentages. Source trail: [S02], [S07], [S08], [S09]

**Q14. Does the absence of an official warning guarantee safety?**

No. Monitoring and communication can be incomplete or delayed. Credible direct signs and immediate danger require protective action and contact with the responsible authority. Source trail: [S02], [S07], [S08], [S09]

**Q15. What information should a follow-up question request?**

Only details that change advice, such as broad locality, time window, relevant slope/river/coastal connection, current signs and whether danger is happening now. Avoid unnecessary personal data. Source trail: [S02], [S07], [S08], [S09]

**Q16. What if maps and observed conditions disagree?**

Do not dismiss immediate danger because a static map is reassuring. Report from safety, follow current official instructions and note map date, scale and uncertainty. Source trail: [S02], [S07], [S08], [S09]

**Q17. What if current alerts cannot be accessed?**

State explicitly that live alerts were not checked, provide conservative general safety guidance and direct the user to the relevant official source or emergency service. Source trail: [S01], [S02], [S07]

**Q18. Can a photograph prove that a site or structure is safe?**

No. Images may omit key features and cannot substitute for site inspection, qualified assessment or live situational information. Source trail: [S01], [S02], [S07]

# Frequently asked questions (continued)

Document 01 | Questions 19-24

**Q19. Where can the public find authoritative information?**

Start with the relevant official Indian/state/district authority and technical agency listed in the source register; verify the publication date, geographic coverage and current status. Source trail: [S01], [S02], [S07]

**Q20. Can old case studies be used as a forecast?**

No. Case studies describe past conditions and lessons, not the exact timing or outcome of a future event. Source trail: [S01], [S02], [S07]

**Q21. What should users avoid doing?**

Do not enter or approach dangerous areas, cross floodwater, inspect unstable slopes, enter damaged buildings or exceed volunteer training. Follow official instructions. Source trail: [S01], [S02], [S07]

**Q22. How should conflicting official messages be handled?**

Compare issuing agency, timestamp, geography and scope. Follow the current directly applicable emergency instruction while seeking clarification without delaying life safety. Source trail: [S01], [S02], [S07]

**Q23. Can the chatbot confirm that emergency services were called?**

Only when an authorized connected service actually confirms the action. Otherwise it should direct the user to call the service directly. Source trail: [S01], [S02], [S07]

**Q24. Are the PDFs a live monitoring system?**

No. They are static reference documents. Current forecasts, alerts, road closures and shelter availability require a verified live integration or direct official check. Source trail: [S01], [S02], [S07]

# Glossary and interpretation rules

Hazard: A potentially damaging process, phenomenon or human activity.

Exposure: People, infrastructure, housing, production capacity and other tangible human assets in hazard-prone areas.

Vulnerability: Physical, social, economic or environmental conditions that increase susceptibility to harm.

Susceptibility: Relative tendency of an area or system to experience a process; not a forecast of immediate occurrence.

Capacity: Resources, systems, relationships and abilities available to manage a hazard and reduce harm.

Forecast: A statement about possible future conditions over a specified place and time, based on data and a method.

Official warning: A communication from a responsible authority describing a hazard and recommended action for stated coverage and validity.

Evacuation: Protective movement away from a threatened area by an appropriate route.

Residual risk: Risk remaining after prevention or mitigation; never automatically zero.

Uncertainty: Limits in data, model assumptions, observations, timing, geography or inference.

RAG: Retrieval-Augmented Generation: retrieval supplies source passages that help ground a model response.

Source provenance: Information identifying the origin, issuer, date, version and context of a claim.

Site-specific assessment: Evaluation using local conditions, current evidence and the competence needed for the decision.

# References and source notes

Document CDVH-01-KB | Version 1.1.0 | Access date 2026-10-16. Official sources may change; re-check currentness before operational use. The source list is not a claim that every linked attachment was completely extracted and reviewed.

[S01] Kerala State Disaster Management Authority guidelines index - KSDMA. Publication/revision date: Index includes 2026 Orange Book of Disaster Management, Edition 8 (individual attachment date must be checked). URL: https://sdma.kerala.gov.in/guidelines/. Accessed 2026-10-16.

Use and limitation: Current index lists the 2026 Orange Book and other guidance; individual linked documents require independent version review.

[S02] Kerala State Disaster Management Plan index - KSDMA. Publication/revision date: Not confirmed during this build. URL: https://sdma.kerala.gov.in/disaster-management-plans/. Accessed 2026-10-16.

Use and limitation: Index page; plans may be older than the current operational situation.

[S07] Disaster risk terminology - UN Office for Disaster Risk Reduction. Publication/revision date: Not confirmed during this build. URL: https://www.undrr.org/terminology/disaster-risk. Accessed 2026-10-16.

Use and limitation: Terminology supports hazard, exposure, vulnerability and capacity framing.

[S08] Exposure - UN Office for Disaster Risk Reduction. Publication/revision date: Not confirmed during this build. URL: https://www.undrr.org/terminology/exposure. Accessed 2026-10-16.

Use and limitation: Reference for people, infrastructure and assets located in hazard-prone areas.

[S09] Vulnerability - UN Office for Disaster Risk Reduction. Publication/revision date: Not confirmed during this build. URL: https://www.undrr.org/terminology/vulnerability. Accessed 2026-10-16.

Use and limitation: Reference for physical, social, economic and environmental susceptibility to harm.

[S10] Sendai Framework for Disaster Risk Reduction 2015-2030 - UN Office for Disaster Risk Reduction. Publication/revision date: Not confirmed during this build. URL: https://www.undrr.org/publication/sendai-framework-disaster-risk-reduction-2015-2030. Accessed 2026-10-16.

Use and limitation: International policy framework; not a substitute for domestic law or local operating procedures.

[S20] Emergency Response Support System 112 - Government of India. Publication/revision date: Not confirmed during this build. URL: https://112.gov.in/. Accessed 2026-10-16.

Use and limitation: Official nationwide emergency response portal; test/verify local access practices before publishing local workflows.

## Citation and update policy

Source identifiers are printed near relevant content and map to this list. For operationally consequential claims, verify the original source, its current status, publication/revision date, geography and relevant section. Do not invent page numbers or infer that a linked attachment was reviewed where the build record says it was unavailable.

## Research limitation

**This is an educational draft reference corpus, not a formally certified emergency manual. Some official web pages were inspected, but not every listed attachment was downloaded and fully extracted. The source register in this PDF records known limitations. Expert review remains necessary for medical, engineering, geotechnical, GIS, legal and operational content. Static PDFs do not provide live alerts or site-specific predictions.**